Laporan Tugas 2

EDA pada Dataset QA arxiv-astro-ph

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1. Analisis banyak data dan missing value
   Pertama saya melakukan cek berapa banyak baris dan kolom pada dataset, dan didapat hasil seperti ini.

```
dtrain = ds["train"]
data = dtrain.to_pandas()[["question", "answer"]]
print(data.shape)
display(data.head(5))
```

```
(10356, 2)
```

	question	answer
0	What criteria were used to isolate young stell...	The researchers used combined IRAC and MIPS c2...
1	What evidence supports the presence of disc ov...	The study observed spin-modulated radial veloc...
2	How can millimeter wavelength observations hel...	Millimeter wavelength observations, particular...
3	How does the theoretical framework explain ene...	The framework posits that energy injected at t...
4	How does the Markov Chain Monte Carlo approach...	The Markov Chain Monte Carlo approach not only...

   Selanjutnya dilakukan cek missing value dan bagusnya di dataset ini tidak ada missing value.

```
data.isnull().sum()
```

	0
question	0
answer	0

dtype: int64

2. Analisis Baris Duplikat
   Selanjutnya saya melakukan cek terhadap baris yang duplikat dan ternyata ada 1 baris yang merupakan duplikat.

```
clean = data.assign(
    question=data["question"].str.strip().str.lower(),
    answer=data["answer"].str.strip().str.lower()
)
dups_clean = clean[clean.duplicated(["question","answer"], keep=False)]
display(dups_clean.sort_values(["question", "answer"]).head(20))
```

	question	answer
219	what correlations were found between infrared ...	the research confirmed a correlation between i...
1977	what correlations were found between infrared ...	the research confirmed a correlation between i...

Jadi saya pun mengambil Langkah untuk menghapus baris yang duplikat tersebut

```
data_dedup = data.drop_duplicates(subset=["question", "answer"], keep="first")
data = data_dedup.reset_index(drop=True)
```

```
clean = data.assign(
    question=data["question"].str.strip().str.lower(),
    answer=data["answer"].str.strip().str.lower()
)
dups_clean = clean[clean.duplicated(["question","answer"], keep=False)]
display(dups_clean.sort_values(["question", "answer"]).head(20))
```

question	answer

3. Analisis Banyak Kata pada masing – masing kolom
   Pada kolom question, didapat top 3 kata yang sering muncul yaitu:

Question Word Frequent

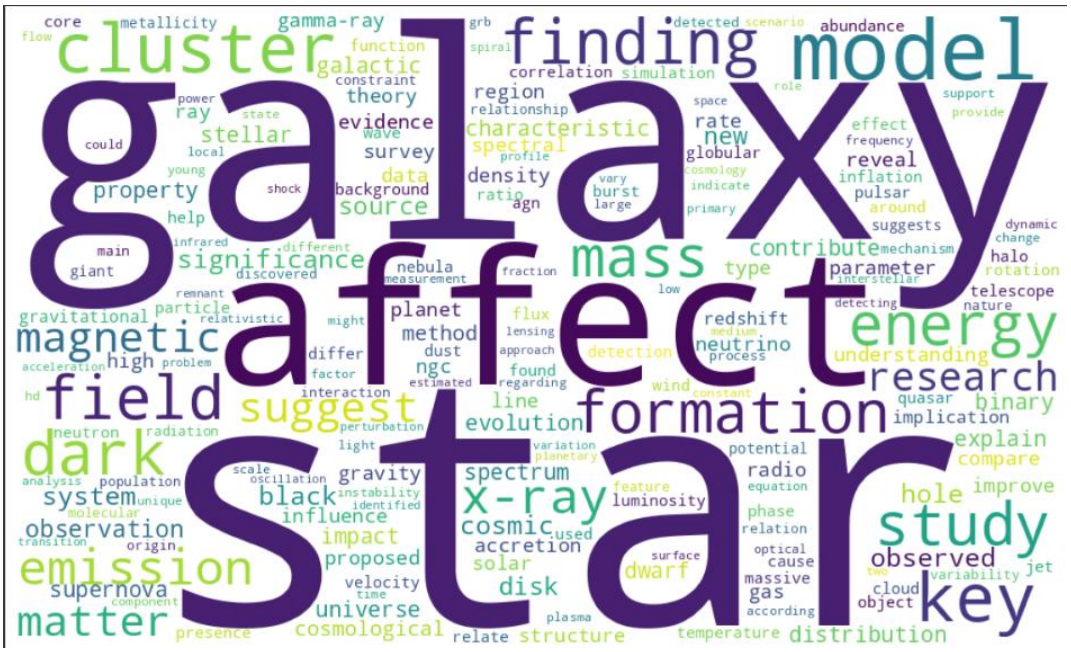
	term	count
0	affect	1183
1	galaxies	695
2	stars	671

Dan pada kolom answer kata yang sering muncul nya yaitu:

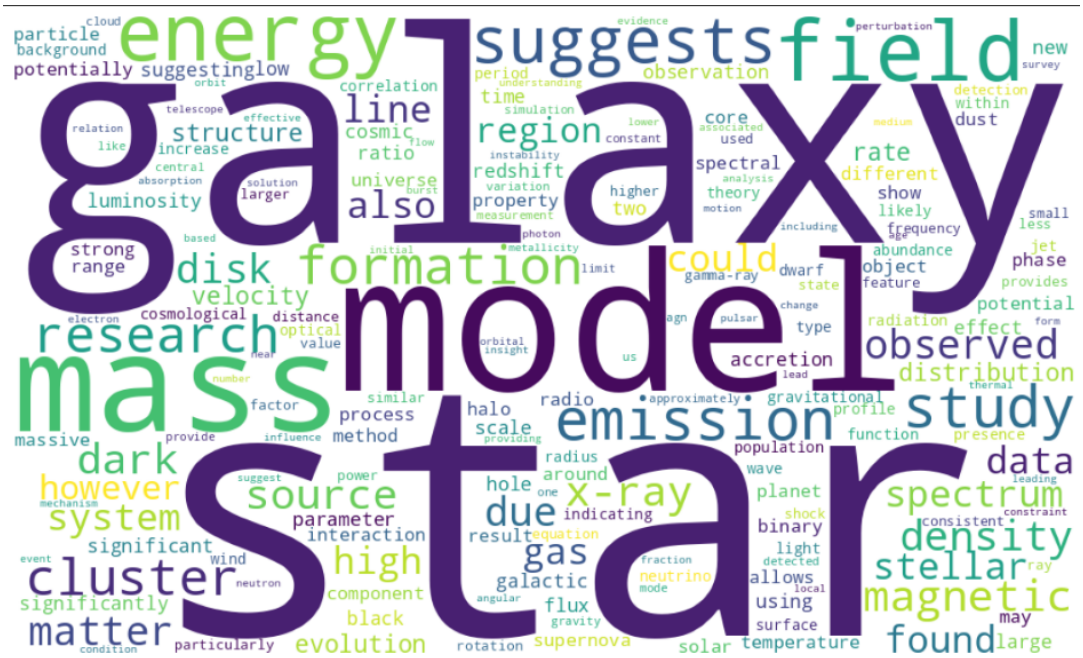
Answer Word Frequent

	term	count
0	mass	1782
1	suggests	1667
2	stars	1638

Disini saya juga membuat visualisasi dengan wordcloud dan didapat hasil seperti ini
Untuk kolom question (Setelah di lemmatizer):



Untuk kolom answer (Setelah di lemmatizer):


galaxy
star
model
mass
energy
suggests
field
formation
research
emission
study
observed
line
region
structure
disk
also
could
dark
however
system
cluster
matter
source
high
gas
due
x-ray
spectrum
density
stellar
magnetic
found
data
distribution
accretion
velocity
cosmological
redshift
property
core
rate
spectral
theory
show
likely
frequency
abundance
object
phase
potential
radiation
effect
type
radio
gravitational
profile
function
population
wave
planet
shock
binary
light
allows
using
surface
temperature
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rotation
evidence
perturbation
new
survey
detection
within
dust
medium
lower
constant
used
higher
two
small
less
jet
provides
form
pulsar
agn
change
including
age
dwarf
gamma-ray
state
feature
limit
metallicity
photon
measurement
variation
instability
period
understanding
time
simulation
observation
correlation
cosmic
ratio
universe
burst
property
measurement
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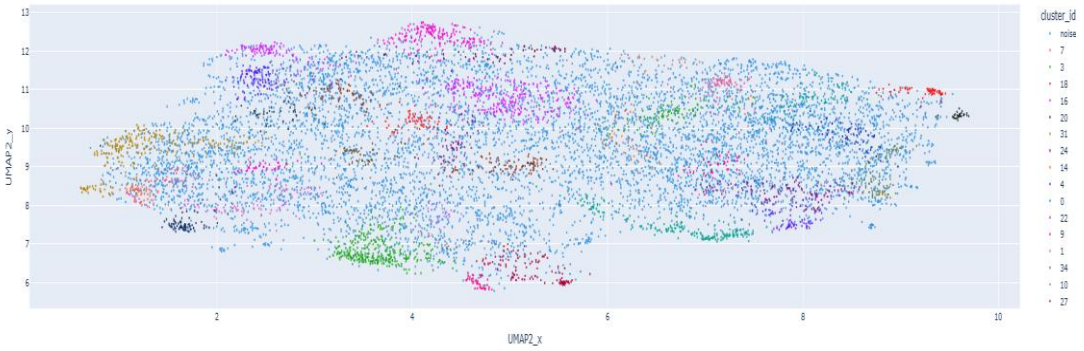
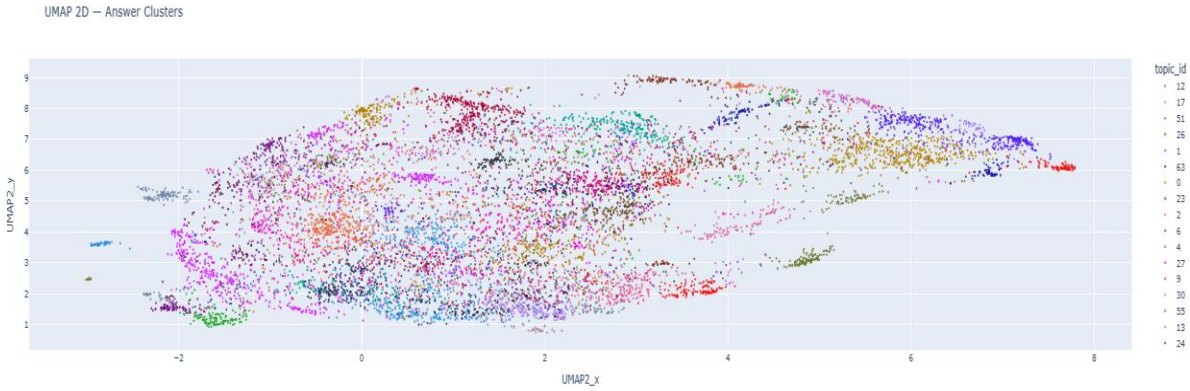
high
gas
galactic
angular
rotation
supernova
flux
gravity
neutrino
fraction
mode
result
equation
due
hole
one
indicating
around
radius

4. Clustering pada kolom question berdasarkan tipe pertanyaan dan topiknya
   Untuk hasil clustering nya didapat visual seperti ini:



   Terlihat disana kalau pada baris question banyak yang tergolong kedalam noise karena kurangnya relasi antar kalimat di question.

5. Clustering pada kolom answer berdasarkan topik yang dibawa
   Selanjutnya yaitu clustering pada kolom answer dengan hasil seperti ini:



   Pada clustering di kolom answer, kita mendapat sekitar 64 kelompok yang berbeda beda dengan kelompok yang paling banyak yaitu yang membahas “cosmological, gravity, universe, scalar, constant, the, quantum, of, theory, general”.